mHC

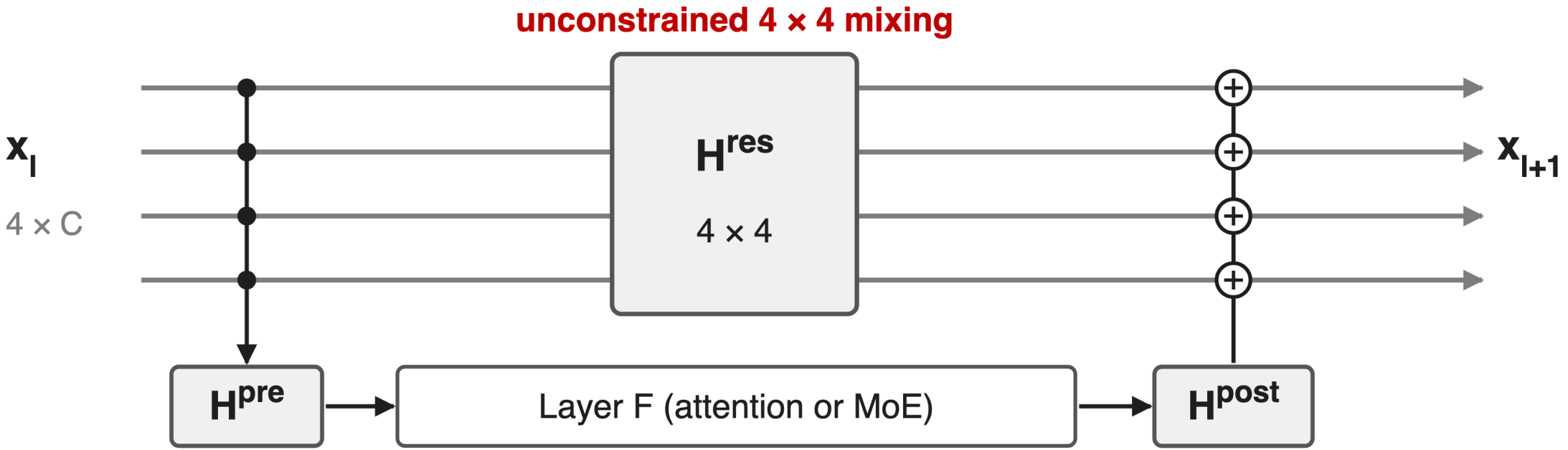
Manifold-constrained hyper-connections keep a 4-lane residual stable

DeepSeek-AI · mHC paper, used in DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Model Shape & Depth Scaling · arxiv.org/abs/2512.24880

Hyper-Connections widen the residual into 4 lanes



The 4 × 4 lane mixing carries most of HC's gain, at almost no FLOPs

Unconstrained mixing amplifies signals ~3000×

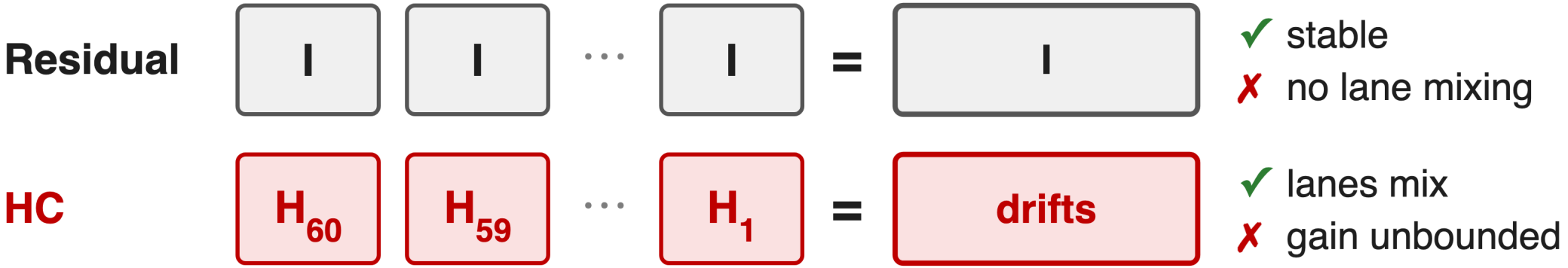
~3000×

peak gain of HC's composite residual mapping in a 27B MoE (ideal: 1); the training loss surges near step 12k

mHC, arXiv 2512.24880, §3.1, Fig. 2 and Fig. 3(b) (27B model, n = 4)

Identity is stable; free mixing compounds

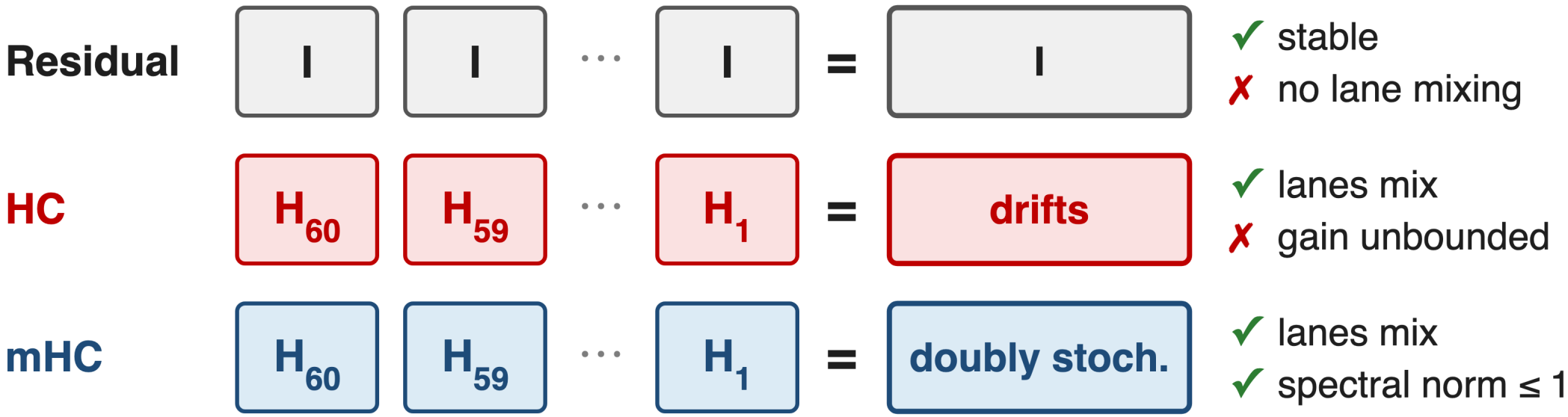
product of residual mappings across depth (60 sublayers in the 27B model)



HC also moves 34C vs 3C residual elements per token, and 4× pipeline traffic

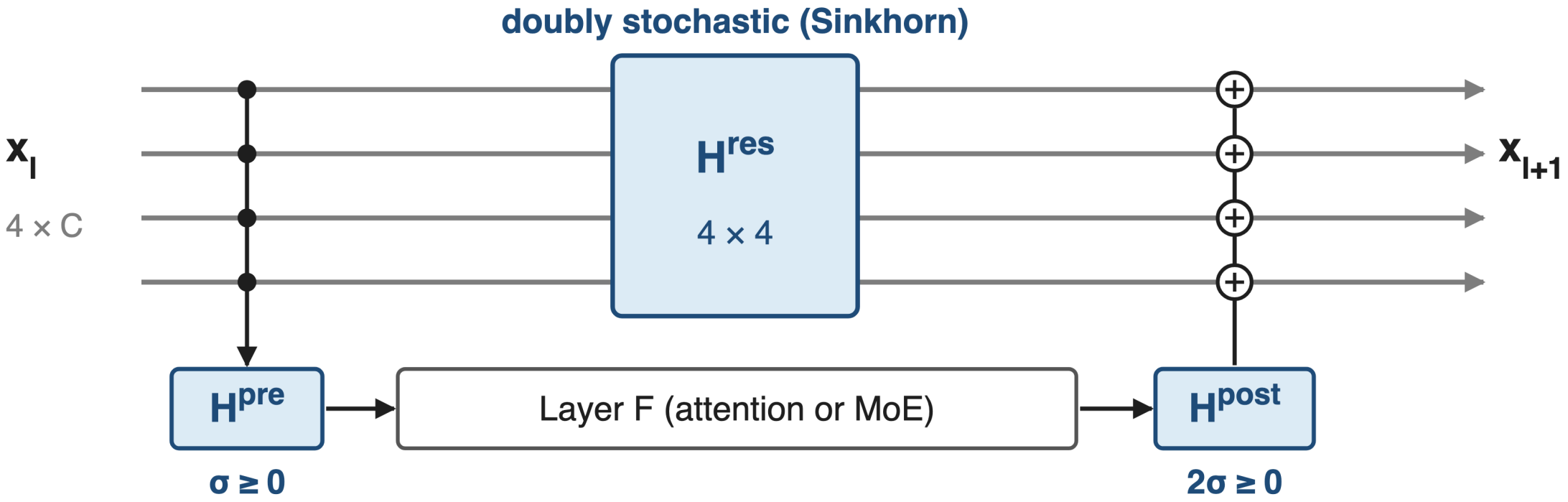
Insight: doubly stochastic products stay bounded

product of residual mappings across depth (60 sublayers in the 27B model)



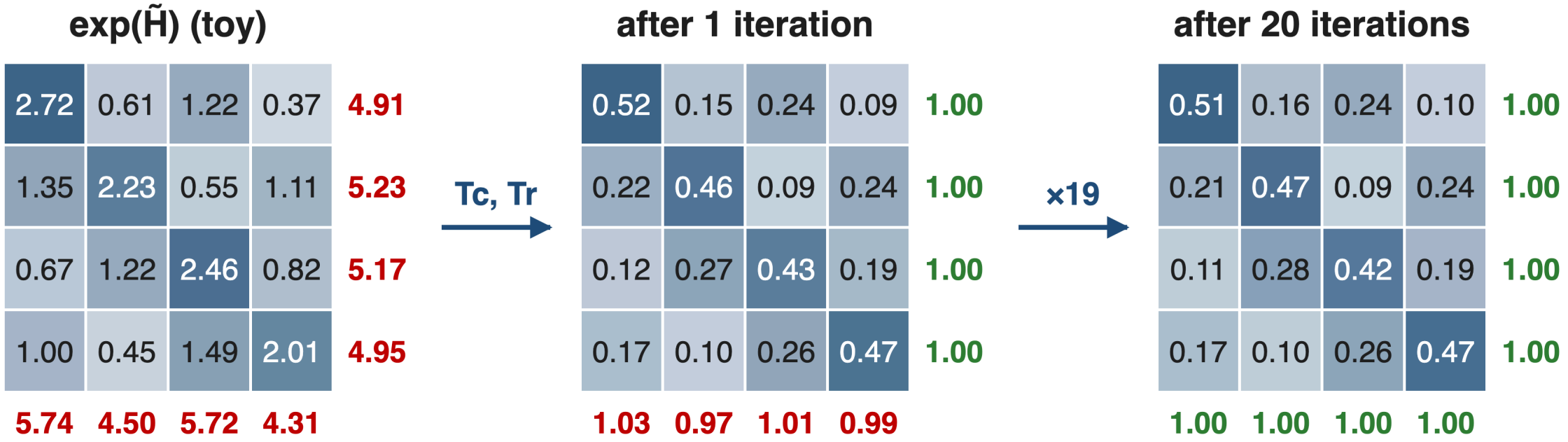
Rows and columns sum to 1: lanes still mix, but no product can amplify

mHC projects each mapping onto a safe set



V4: every block, n = 4 lanes, 20 Sinkhorn iterations per mapping

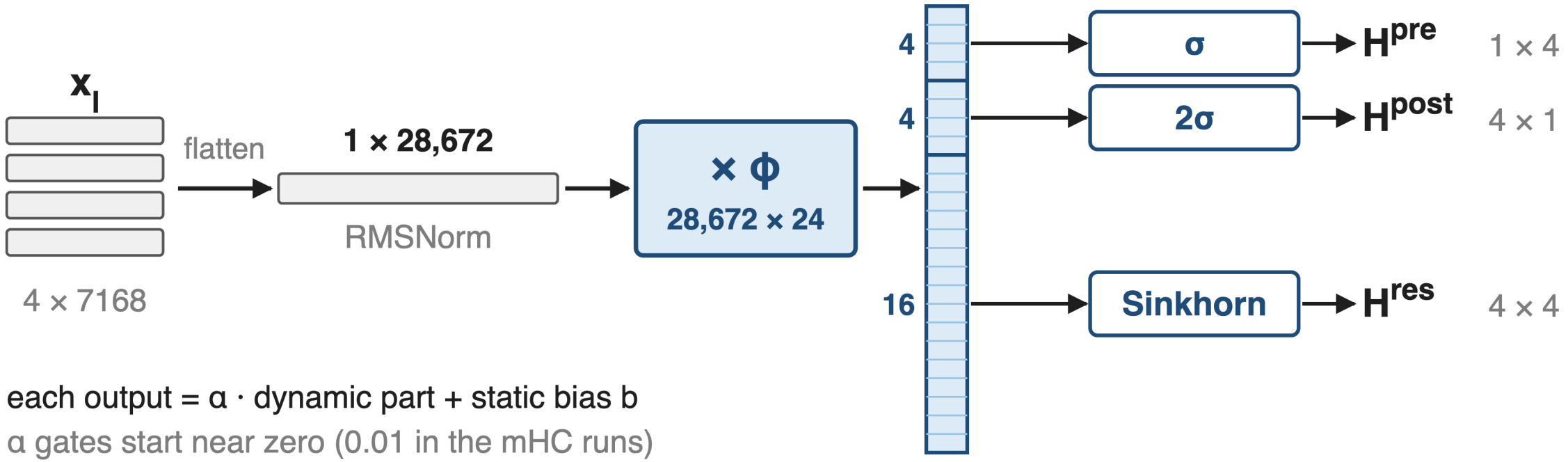
❶ Sinkhorn-Knopp balances rows and columns



right: row sums · below: column sums · red ≠ 1, green = 1

exp makes entries positive; alternating column and row scaling converges

❷ One 24-wide projection drives all mappings



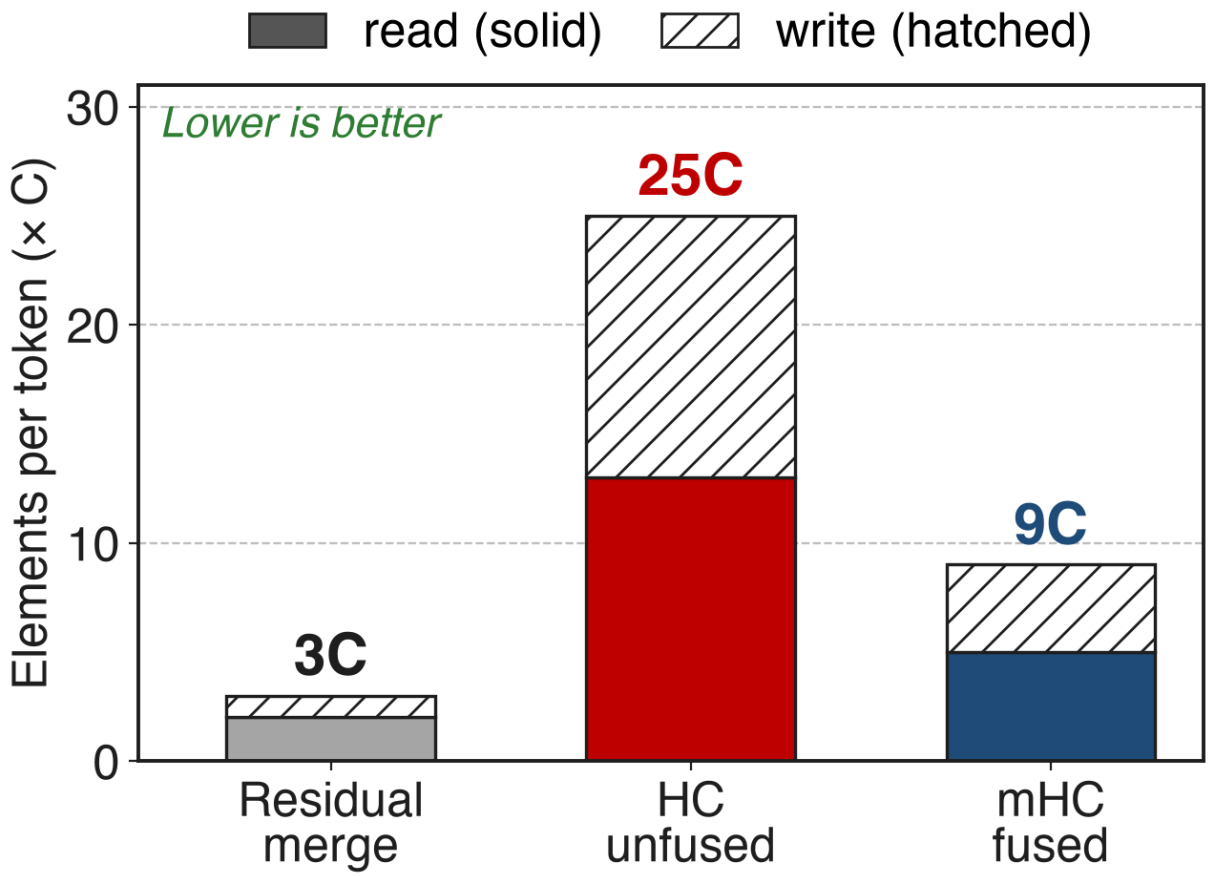
each output = α · dynamic part + static bias b

α gates start near zero (0.01 in the mHC runs)

Mixing depends on the token, for a 28,672 × 24 projection in V4-Pro

❸ Fused kernels cut merge traffic 25C → 9C

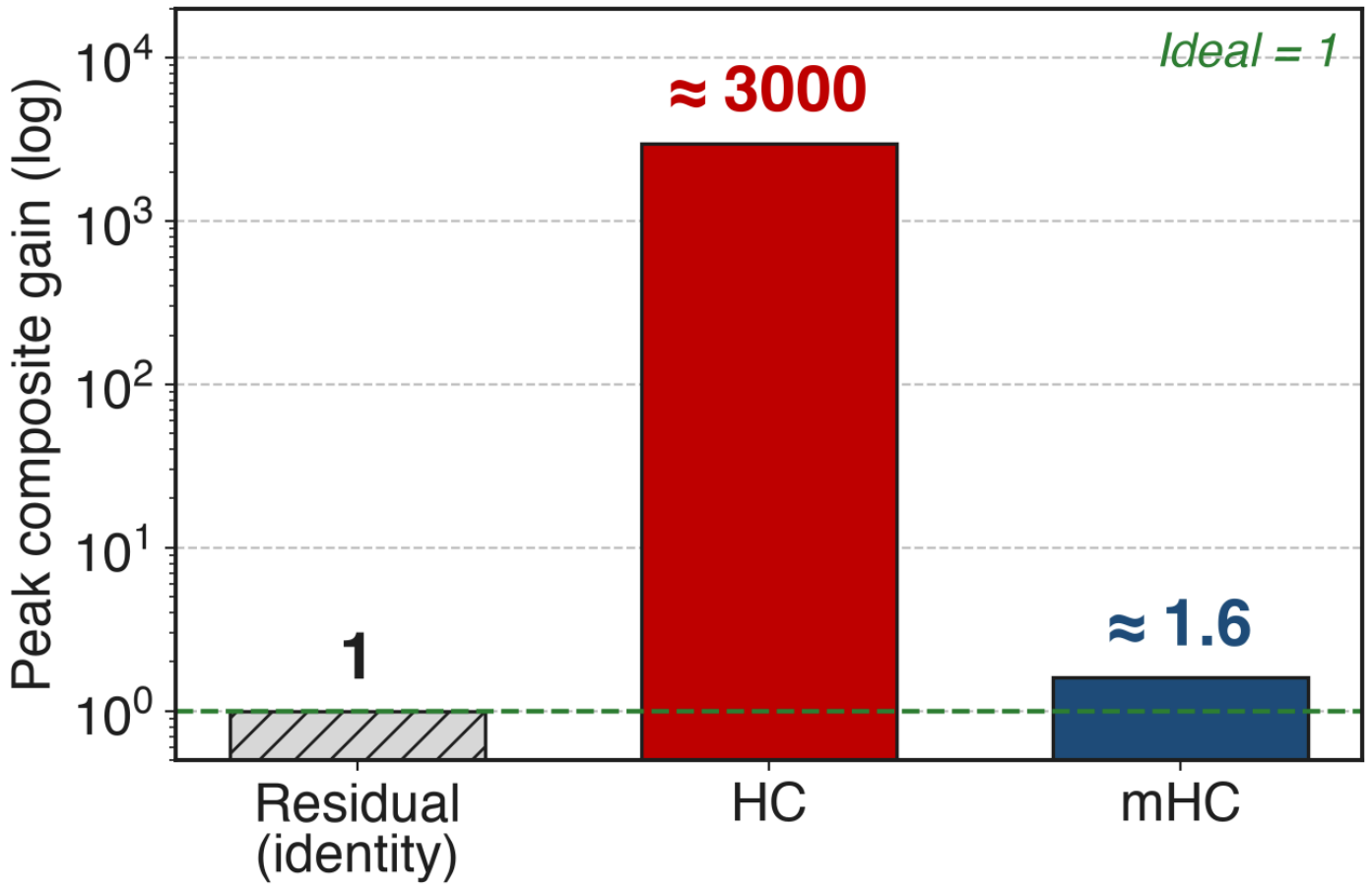
- One kernel: post-mapping, lane mixing, merge
- Recompute mHC; store inputs at pipeline-stage boundaries
- DualPipe runs the merge on a high-priority stream



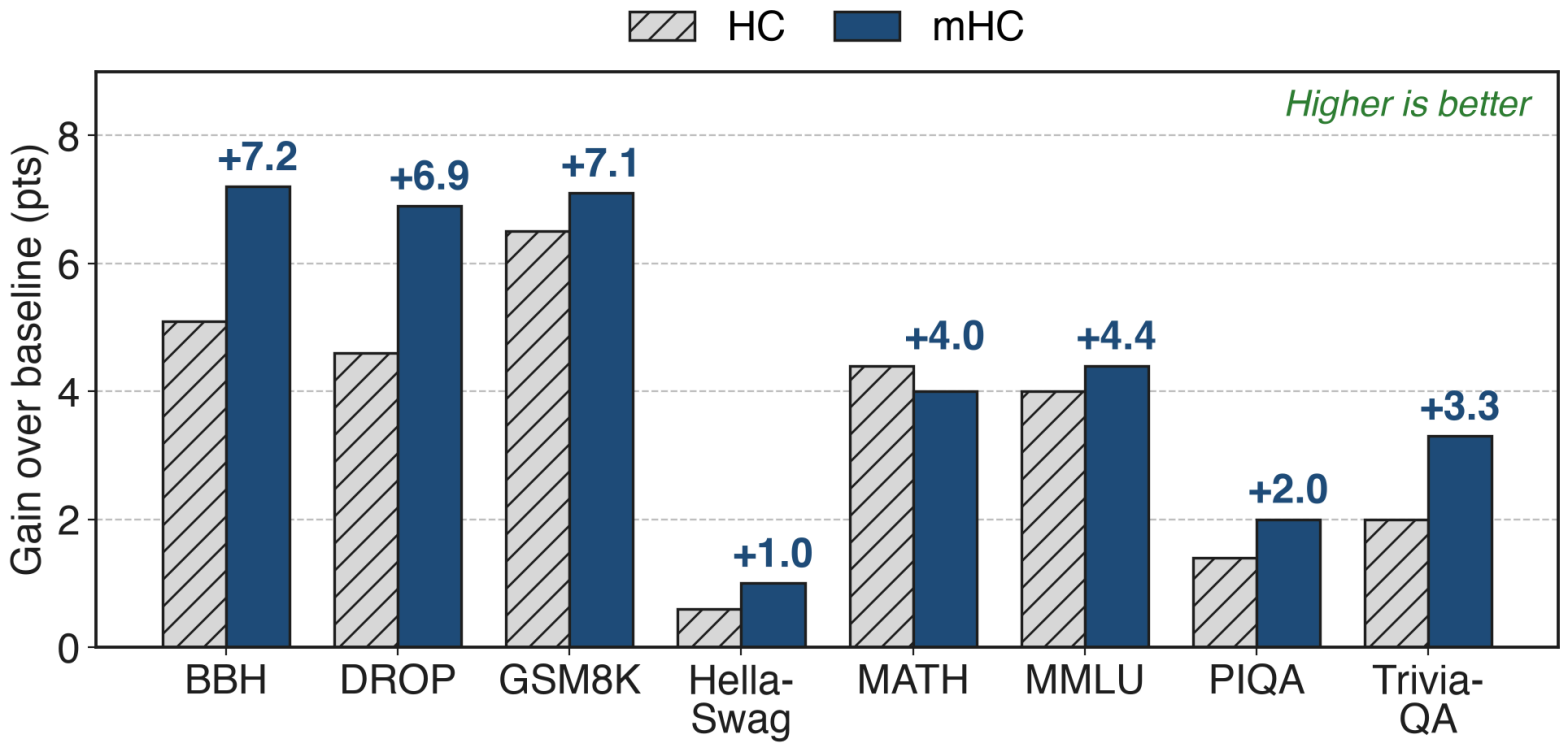
Analytic, n = 4, per token, write-back step only

Composite gain drops from ~3000 to 1.6

- 27B MoE, n = 4, 20 Sinkhorn steps
- No loss surge; gradient norm tracks baseline
- Final loss **–0.021** vs baseline



mHC beats the baseline on all eight benchmarks



27B: BBH +7.2, DROP +6.9 over baseline; ahead of HC on 7 of 8

The cost: 6.7% step time, custom kernels

- **6.7%** more training time at n = 4
- **4×** residual activations and stage traffic remain

- 20 Sinkhorn steps only approximate the constraint
- Needs fused TileLang kernels (deck 41)
- Evidence: DeepSeek in-house runs only

Summary

- HC's free lane mixing: composite gain **~3000×**, loss surges
- **Insight**: doubly stochastic products cannot amplify
- Sinkhorn (**20** steps), σ / 2σ, fused kernels; **n = 4** in V4
- Gain ≈ **1.6**, loss **–0.021**, BBH **+7.2**; **6.7%** overhead

https://arxiv.org/abs/2512.24880